



Review

How three linked clinical observations led to an understanding of perioperative heat balance: A personal reflection on the scientific process

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HIGHLIGHTS

- The rapid decrease in core temperature after induction of general anesthesia led to an understanding of redistribution hypothermia.
- The linear reduction in core temperature during the subsequent 2–3 h led to an understanding of metabolic heat production and cutaneous heat loss.
- The core temperature plateau at about 34.5 °C led to the understanding that anesthetized patients are not poikilothermic, and that arteri-venous shunt constriction constrains metabolic heat to the core thermal compartment.

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ABSTRACT

Three linked clinical observations prompted our current understanding of perioperative heat balance. The first was the extraordinarily rapid decrease in core temperature after induction of general anesthesia which led to an understanding of redistribution hypothermia. The second was the linear reduction in core temperature during the subsequent 2–3 h which led to an understanding of anesthetic effects on metabolic heat production and factors that influence cutaneous heat loss. And the third was the observation that core temperature reaches a plateau at about 34.5 °C which led to the understanding that thermoregulatory vasoconstriction re-emerges when patients become sufficiently hypothermic, and that arteri-venous shunt constriction constrains metabolic heat to the core thermal compartment.

In 1985, I was recruited to the University of California San Francisco (UCSF) by Ronald Miller directly from my pediatric and anesthesia residencies. Ron needed a pediatric anesthesiologist who was also a chemist who could help George Gregory and Laurence Litt establish a magnetic resonance spectroscopy research team. (Ron was mistaken in believing I was a chemist. The chemist in our family is my brother Jonathan, now a member of the National Academy of Sciences — as was my father.) I spent about a year studying magnetic resonance spectroscopy [1] before returning to thermoregulatory research which I had started in a desultory way as a resident at the University of California Los Angeles. The near-universal opinion in the Department was that I was leaving real science to mess around with thermometers.

My first thermoregulatory project was building a physiological data-acquisition system (Fig. 1). The hardest part was learning to program LabVIEW (National Instruments, Austin, Tx), a special type of data-driven software that was the best available for handling real-time data. For four months, aside from my clinical Tuesdays, I programmed at home. Somewhat remarkably, not a single person asked why I didn't

come to the office or what I was doing with my time. UCSF was that kind of place.

The thermoregulatory data-acquisition system measured everything necessary for sophisticated thermoregulatory studies including electromyograms, oxygen consumption, arterio-venous shunt tone (various ways), sweating, core temperature, average skin temperature, heat flux, ambient temperature and humidity, etc. All signals were analyzed in real time to guide study progress. My investment in the system proved worthwhile as we used it for more than 100 studies.

1. Redistribution hypothermia

Soon after arriving at UCSF, I anesthetized a normothermic adult kidney donor. Anesthetic induction took perhaps 10 min. I then had the chance to insert an esophageal temperature probe which showed a core temperature near 35.5 °C. It was beyond me how someone could lose so much heat in just a few minutes. If that much heat had been dissipated to the environment, my patient would have been glowing like a light bulb.

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Fig. 1. Dr. Sessler's thermoregulatory data-acquisition system, integrated with a Dräger anesthesia machine. The thermoregulatory data-acquisition system measured everything necessary for sophisticated thermoregulatory studies including electromyograms, oxygen consumption (as a measure of heat production), arterio-venous shunt tone (various ways), sweating, core temperature, average skin temperature (15 sites), heat flux (15 sites), ambient temperature and humidity, etc. All signals were analyzed in real time to guide study progress.

The observation was simply inconsistent with my understanding of heat balance and tissue temperature. I mentioned the problem to Larry Litt (one of two UCSF anesthesia faculty who had PhDs in physics). He gave the same answer that my parents (both physicists) presumably would have given: “thermodynamics is never wrong; it is always the assumptions that are wrong.”

Larry was right, of course. His insightful comment led me to understand that heat is not evenly distributed in the body. As is now well-understood, hypothermia after induction of anesthesia results from a large internal core-to-peripheral redistribution of body heat rather than dissipation of heat to the environment. A detailed understanding of heat flow during induction of general [2] and epidural [3] anesthesia would have to wait almost 10 years while we developed an understanding of how anesthesia impairs thermoregulation [4,5] and methodology for quantifying regional heat content in human volunteers. That led to a key study of heat flow within the body, heat loss to the environment, and regional tissue temperature during induction of general anesthesia [2].

Essential background is that while general anesthesia has little effect on warm defenses, it profoundly impairs thermoregulatory arterio-venous shunt vasoconstriction [4–7]. Any dose of general anesthesia thus completely relaxes tonic arterio-venous shunt tone that normally maintains the core-to-peripheral thermal gradient. Along those lines, there is a common assumption that veins “pop out” on the back of the hand after induction of anesthesia because anesthetics are direct vasodilators. In fact, vasodilation largely results from centrally mediated relaxation of tonic thermoregulatory arterio-venous shunts in the fingers.

Another important bit of background is that there is normally a core-to-peripheral temperature gradient. Most metabolic heat is produced in

the core. The Second Law of Thermodynamics specifies that heat can only flow down a temperature gradient. To prevent heat from accumulating in the core, there must therefore on average be a thermal gradient from the core to peripheral tissues, to skin, and finally to the environment. The thermal core (highly perfused tissue of the trunk and head) is about the same size as the peripheral thermal compartment, mainly the arms and legs.

Heat loss to the environment, while appearing to be the obvious mechanism, is not the cause of the initial rapid reduction in core temperature after induction of anesthesia. Instead, hypothermia results from dilation of thermoregulatory arterio-venous shunts which allows heat to flow from the core to the periphery — thus reducing core temperature far out of proportion to the small reduction in overall body heat content (Fig. 2). Both intravenous [5] and volatile [4] anesthetics profoundly impair cold defenses; consequently, the magnitude of redistribution hypothermia is comparable with each. Redistribution also explains the hypothermia that occurs shortly after induction of neuraxial anesthesia although the magnitude is somewhat less since vasoconstriction normally remains intact in the arms [3]. A corollary is that warming operating rooms during induction has no appreciable effect on core temperature [8].

Redistribution hypothermia is hard to treat because the internal flow of heat is large; furthermore, heat applied to the skin surface takes time to reach the core. However, redistribution hypothermia can be prevented by pre-warming [9]. Pre-warming does not much change core temperature which remains regulated, but increases peripheral compartment temperature, thus reducing the core-to-peripheral tissue temperature gradient. Because heat flow is proportional to thermal

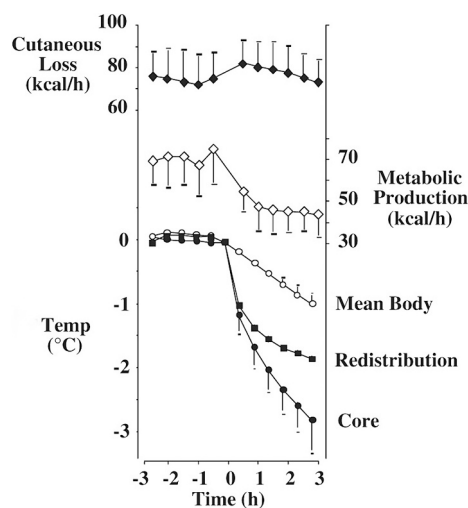


Fig. 2. Changes in body-heat content and distribution of heat within the body during induction of general anesthesia. Mean body temperature is the average temperature of all tissues including the arms and legs, whereas core temperature is the temperature of the highly perfused tissues of the trunk and head. Heat loss and metabolic heat production were initially similar. Overall heat balance was thus near zero before induction of anesthesia (at elapsed time zero), but subsequently decreased ≈ 31 kcal/h. The contributions of decreased overall heat balance and internal redistribution of body heat to the decrease in core temperature were separated by multiplying the change in overall heat balance by body weight and the specific heat of humans. The resulting change in mean body temperature (“mean body”) was subtracted from the change in core temperature (“core”), leaving the core hypothermia specifically resulting from redistribution (“redistribution”). After 1 h of anesthesia, core temperature had decreased 1.6 ± 0.3 °C, with redistribution contributing 81% to the decrease. During the subsequent 2 h of anesthesia, core temperature decreased an additional 1.1 ± 0.3 °C, with redistribution contributing only 43%. Redistribution thus contributed 65% to the entire 2.8 ± 0.5 °C decrease in core temperature during the 3 h of anesthesia. All results are shown as means $\pm$ SDs. From Matsukawa et al. [2], with permission.

gradient, decreasing the gradient reduces redistribution hypothermia.

2. The linear hypothermia phase

During long operations in unwarmed patients, redistribution hypothermia which is largely complete within an hour, is followed by a 2- to 3-h-long linear decrease in core temperature. Why core temperature decreases and why the reduction in core temperature should be linear during this phase was unknown, as were the factors that influence the slope.

The second phase of the typical hypothermia curve in unwarmed patients results from heat loss exceeding heat production, with two mechanisms contributing [10]. The first is a somewhat remarkable $\approx 30\%$ reduction in metabolic heat production during general anesthesia [2]. Metabolic heat production decreases about 6% per degree C of core hypothermia in adults [11] and infants [12], but is otherwise nearly constant during anesthesia and minimally influenced by anesthetic technique. Peripheral blocks do not reduce metabolic heat production, and neuraxial anesthesia has little effect on metabolic heat production. Respiratory heat loss is trivial (5–7%), even with a nearly open circuit and unwarmed dry gases [12]. The second mechanism is cutaneous heat loss which can be substantial. Cutaneous loss is largely determined by surface insulation and ambient temperature, and is therefore largely under our control. Modern warmers largely prevent cutaneous loss [13], thus preventing hypothermia and even rewarming hypothermic patients.

The slope of the linear phase of the hypothermia curve depends on the difference between metabolic heat production and cutaneous and respiratory heat loss. Convection and radiation are the major causes of heat loss, with conduction and evaporation normally contributing little. It also depends on body mass and the average specific heat of human tissue which is about $0.83 \text{ kcal/kg/}^\circ\text{C}$ [14]. During the linear phase, temperature change can be modelled as net heat loss divided by the product of body mass and tissue specific heat. While typically negative, the slope can be positive when heat loss is reduced to below metabolic heat production by high ambient temperature, sufficient insulation, or effective warming systems [15].

3. The core-temperature plateau

During my early days at UCSF, effective patients warming systems were not yet available. Consequently, most surgical patients became hypothermic — a condition that was considered normal at the time. (We now know that cardiovascular outcomes, surgical site infection, and transfusion requirements are similar at 35.5 and 37°C [16], but that lower temperatures delay drug metabolism [17,18], prolong post-operative recovery [19], provoke thermal discomfort and shivering [20], and possibly promote wound infections [21].) But a curious aspect of intraoperative hypothermia was that patients did not simply keep getting colder and colder during prolonged major surgery. Instead, core temperature decreased to about 34.5°C , thereafter remaining nearly constant no matter how large and long the operation (Fig. 3).

Like redistribution hypothermia, thermal equilibrium at 34.5°C was inconsistent with a simple thermodynamic explanation. General anesthesia reduces metabolic rate about 30%, and it thereafter decreases about 6% per degree C reduction in mean body temperature [11]. Reduced heat production is counter-balanced by a small reduction in cutaneous heat loss as skin temperatures becomes slightly cooler as core temperature decreases. But cutaneous heat loss does not decrease enough to provoke thermal equilibration on a whole-body basis at 34.5°C during large operations. Again, there had to be another explanation which led to another key study which used similar heat balance quantification, this time to understand why core temperature reaches a plateau [22].

The key to understanding the core-temperature plateau is that humans are not poikilothermic during anesthesia — as was previously

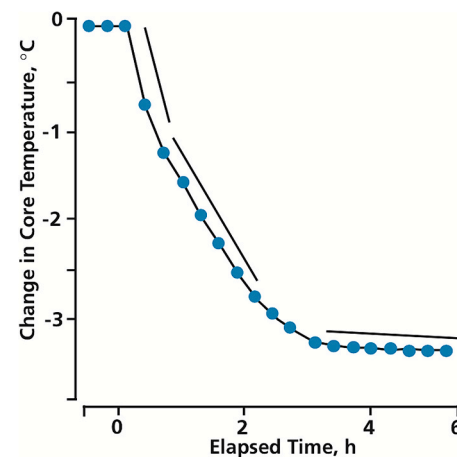


Fig. 3. Hypothermia during general anesthesia develops with a characteristic pattern. An initial rapid decrease in core temperature results from a core-to-peripheral redistribution of body heat. This is followed by a slow, linear reduction in core temperature that results simply from heat loss exceeding heat production. Finally, core temperature stabilizes, and subsequently remains virtually unchanged. This plateau phase may be a passive thermal steady state or result when sufficient hypothermia triggers thermoregulatory vasoconstriction. Results are presented as means $\pm$ SDs. Data from Kurz et al. [21]

assumed. Instead, sufficient hypothermia provokes re-emergence of arterio-venous shunt constriction (and even shivering with sufficient hypothermia). The threshold (triggering core temperature) for vasoconstriction is drug- and dose-dependent. But with typical combinations of anesthetic drugs, vasoconstriction re-emerges at about 34.5°C . Although restricted to fingers and toes, thermoregulatory vasoconstriction effectively constrains metabolic heat to the thermal core, thus preventing further core hypothermia — even though heat balance with the environment remains negative. The core-temperature plateau thus results when re-emergent thermoregulation starts to re-establish the normal core-to-peripheral thermal gradient (Fig. 4).

There is an exception to the general rule that core temperature during general anesthesia reaches a plateau at about 34.5°C . When general and neuraxial anesthesia are combined, core temperature continues to decrease. There are two reasons. The first is that neuraxial anesthesia centrally reduces the vasoconstriction and shivering thresholds [23–25]. And the second is that neuraxial blocks prevent arterio-venous shunt vasoconstriction in the toes, which like all thermoregulatory defenses, are neurally mediated. Re-emergence of thermoregulatory vasoconstriction thus occurs at a lower core temperature when general and neuraxial anesthesia are combined, and is restricted to the fingers which is insufficient to prevent continuing hypothermia [26].

4. Conclusions

In summary, three linked clinical observations led to our current understanding of perioperative heat balance. The first observation was the extraordinarily rapid decrease in core temperature after induction of general anesthesia which led to dozens of thermoregulatory and heat balance studies. Together, they identified the mechanism, namely core-to-peripheral heat flow during and shortly after induction of anesthesia. The clinical correlate was that while redistribution hypothermia is hard to treat, it can be ameliorated by pre-warming.

The second clinical observation was the prolonged linear reduction in core temperature during the second and third hours of anesthesia. One mechanism is that general anesthesia reduces metabolic heat production by a surprisingly high $\approx 30\%$. Simultaneously, cutaneous heat loss remains high, with additional loss from open surgical incisions — the only kind in those days! The consequence is a linear decrease in core temperature that results from heat loss exceeding heat production, with

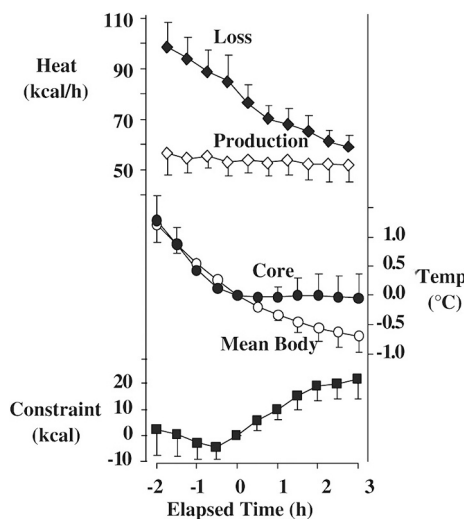


Fig. 4. Changes in body-heat content and distribution of heat within the body during the core-temperature plateau in anesthetized subjects. Mean body temperature is the average temperature of all tissues including the arms and legs, whereas core temperature is the temperature of the highly perfused tissues of the trunk and head. Vasoconstriction decreased cutaneous heat loss ≈ 25 kcal/h. However, heat loss exceeded heat production throughout the study. Consequently, mean body temperature, which decreased at a rate of ≈ 0.6 °C/h before vasoconstriction, subsequently decreased at a rate of ≈ 0.2 °C/h. Core temperature also decreased at a rate of ≈ 0.6 °C before vasoconstriction, but remained virtually constant during the subsequent 3 h. Because mean body temperature and body-heat content continued to decrease, constraint of metabolic heat to the core thermal compartment caused the core-temperature plateau. That is, vasoconstriction reestablished the normal core-to-peripheral temperature gradient by preventing metabolic heat (which is largely generated in the core) from escaping to peripheral tissues. Constrained heat is presented cumulatively, referenced to the onset of intense vasoconstriction defined as time zero; data are expressed as means $\pm$ SDs. From Kurz et al., with permission [22].

the slope determined by the difference. Because the linear phase mostly results from heat loss/gain, the slope is largely determined by the efficacy of passive insulation and/or active warming.

The third clinical observation was the core-temperature plateau during large and long operations. It too led to dozens of studies, and eventually an understanding of how effectively thermoregulatory vasoconstriction constrains metabolic heat to the core thermal compartment. The clinical consequence is that vasoconstriction, once triggered, prevents further intraoperative hypothermia — although patients should not be allowed to get that cold. Our current understanding of perioperative heat balance thus resulted from three linked clinical observations which prompted dozens of explanatory studies.

Junior investigators often ask how to develop strong research questions. The question is pertinent because, assuming studies are technically competent, importance of the work is determined by the quality of the question. Developing clinical research hypotheses is challenging. It is easy to think of important questions that cannot be addressed because of ethical, financial, or practical constraints. Conversely, it is easy to think of studies that address trivial questions. The challenge is to identify meaningful hypotheses that can be addressed in practical studies. There is no simple method, and even experienced investigators consider dozens of ideas before selecting one to attempt. But when searching for research questions, patients are always a good place to look!

CRediT authorship contribution statement

Daniel I. Sessler: Writing – review & editing, Writing – original

draft, Conceptualization.

Declaration of competing interest

Dr. Sessler occasionally consults for temperature-related companies and donates all fees to charity.

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